October 14th, 2026

DATA 4380

# Stress Profiles of STEM College Students in Pakistan



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# Background and Motivation

## The problem

- STEM students face multiple sources of academic and personal stress
- Students may experience stress in different ways
- A single solution may not work for everyone

## Why it matters

- Stress can affect a students wellbeing, focus, and academic performance
- Understanding different stress patterns can help identify a students needs

## Who benefits?

- Pakistani colleges and universities
- Students, counselors, and educators
- Organizations developing better student support systems

## Our motivation

- As STEM students, we understand the relevance of academic stress
- We want to explore how data can help create more supportive learning environments



# Previous Work / Existing Research



### Khan et al. (2013) - Crucial Factors Affecting Stress: A Study Among Undergraduates In Pakistan

- **Data/method:** 188 undergraduates; questionnaire, correlation and chi-square tests.
- **Finding:** 90.4% reported feeling bad about poor grades; questionnaire reliability was (alpha = 0.807).

### Nasir (2025) - Depression, Anxiety, and Stress among University Students in Pakistan

- **Data/method:** 300 university students; DASS-21, Pearson correlation, multiple regression and SEM.
- **Finding:** Academic pressure correlated with stress ($r = 0.49$, $p < 0.01$). Regression explained 41% of stress-score variation ($R^2 = 0.41$).
- **Additional finding:** Coping strategies (beta = -0.33) and peer support (beta = -0.25) were negatively associated with stress in the regression model.

### Saeed et al. (2020) - A Qualitative Exploratory Study of the Factors Causing Academic Stress in Undergraduate Students in Pakistan

- **Data/method:** 80 male students; qualitative research using structured interviews.
- **Finding:** 96.25% experienced academic stress; 90% identified poor time management and 86% cited frequent exams and the semester system.

- **Similarities:** Investigate stress-related factors among university students
- **Differences:** Previous studies used interviews and statistical modeling, we use unsupervised learning to identify distinct stress profiles among STEM students
- **Our contribution:** Analysis of important stress variables, and comparisons between male and female students



# Problem Statement

Can unsupervised learning identify distinct student stress profiles, determine which stress variables contribute most to profile identification, and reveal whether these key variables differ between male and female students?

# Objectives

- Identify stress profiles across selected Pakistani universities.

- Identify which stress relievers are most effective; which stress relievers are common for students with “lower” stress profiles.

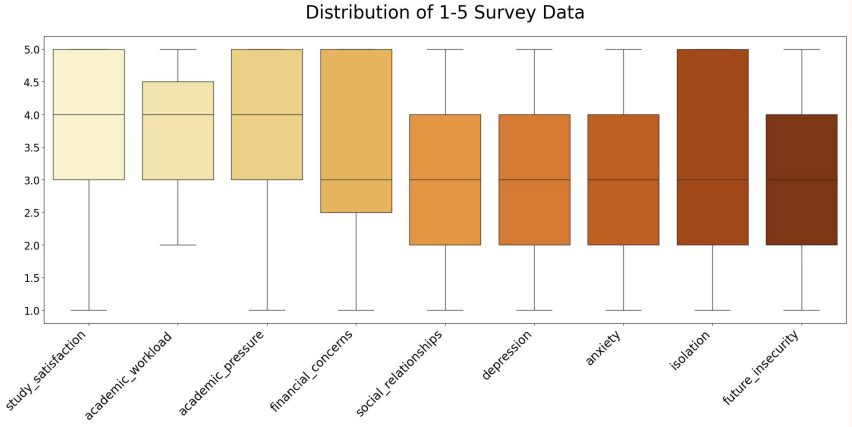
- Identify differences in stress profiles between Male and Female students.

- Identify the variables, other than stress relievers, has the greatest impact on student stress profiles and academic standing.

# EDA & Data Understanding

**1-5 Survey Data**

- Most students report higher level of study satisfaction and academic pressure
- There is a broad range of financial concerns and feelings of isolation, with a median of 3
- Depression and Anxiety have a median of 3

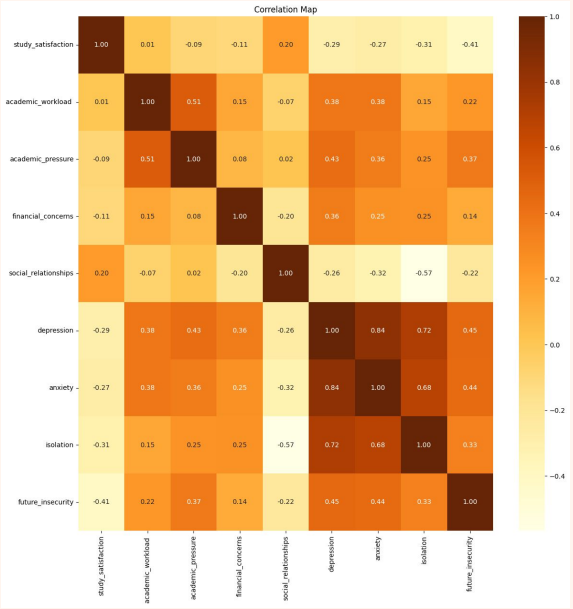


# EDA & Data Understanding

### 1-5 Survey Data

- **Positive Correlation**
  - Depression and Anxiety (0.84)
    - Students reporting higher depression scores tend to report higher anxiety scores
  - Depression and Isolation (0.72)

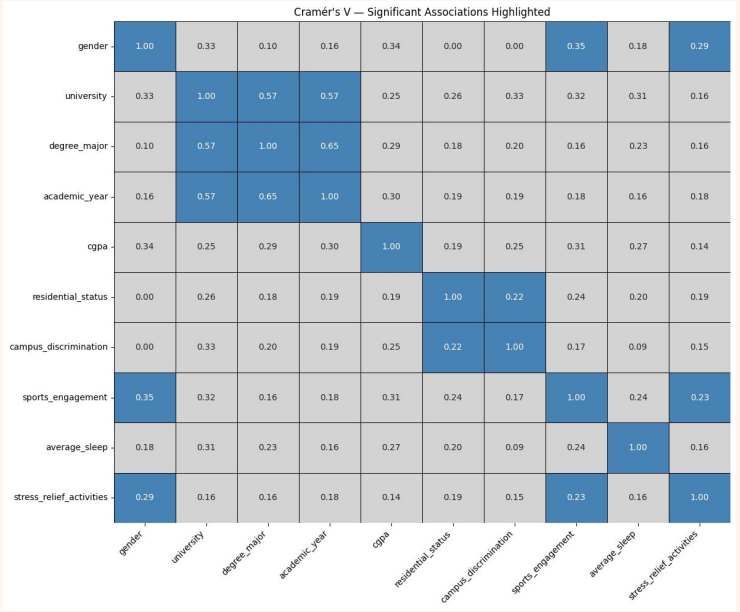
- **Negative Correlation**
  - Study Satisfaction and Future Insecurity (-0.41)
  - Social Relationships and Isolation (-0.57)
    - The higher a student rates their quality of social relationships, the lower they feel isolated.



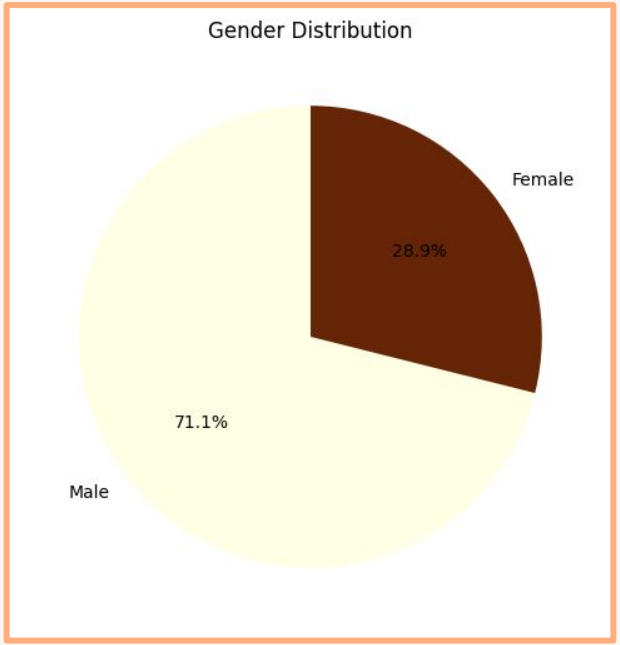
# EDA & Data Understanding

## Categorical Data

- **Correlations using Cramer's V Association**
  - Academic Year and Degree Major (0.65)
  - Gender shows association with sports_engagement (0.35), cgpa (0.34), and stress_relief_activities (0.29)

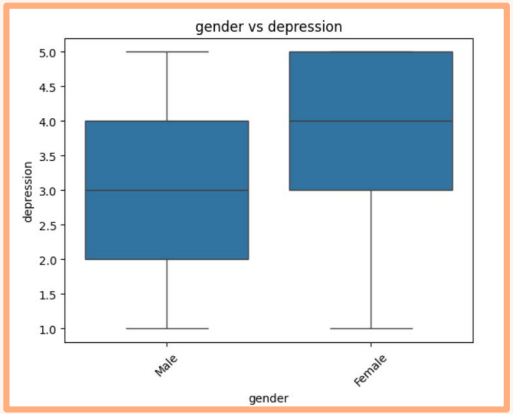


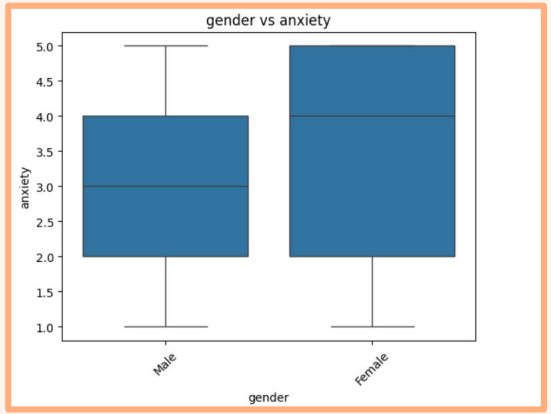
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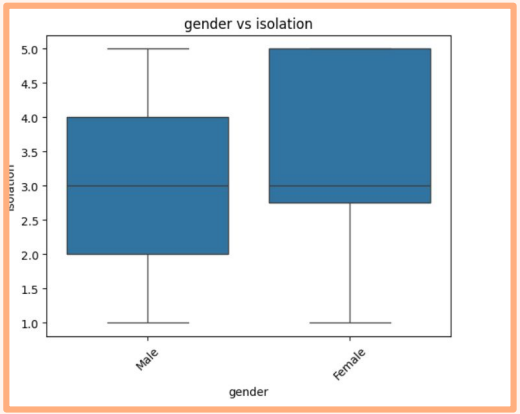


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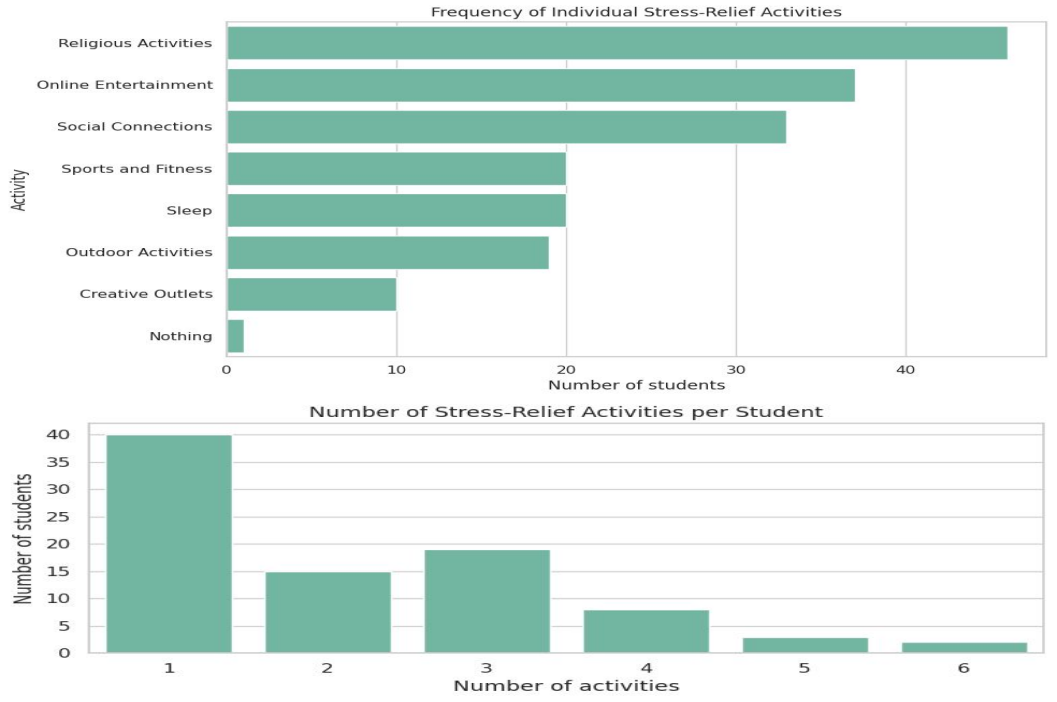
## Gender vs. Mental Health Indicators



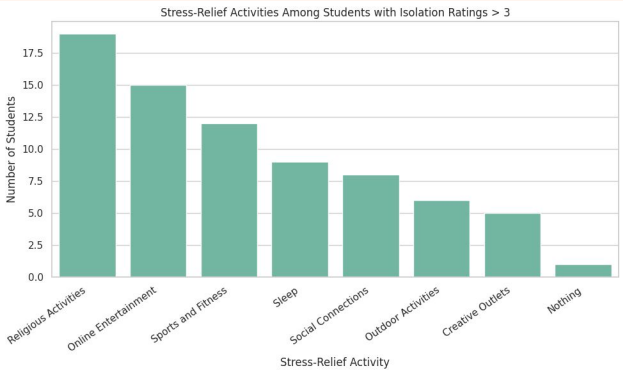
# EDA & Data Understanding

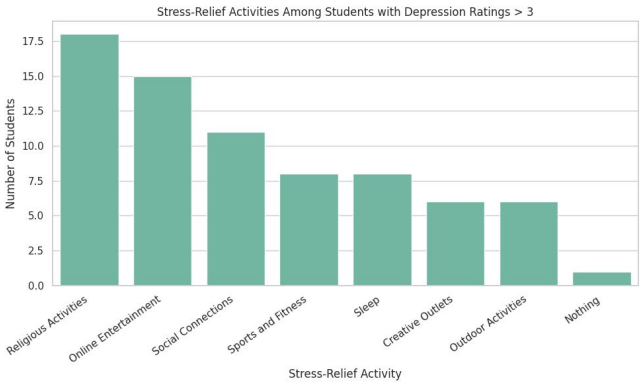


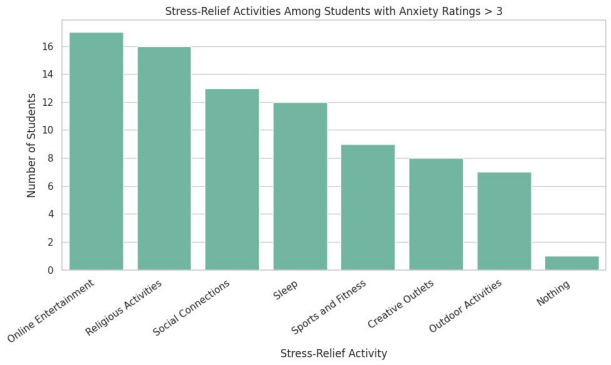

**Stress Relief Activities:**

- Most students list Religious Activities as a stress relief activity, as well as Online Entertainment and Social Connections.
- One student list having no stress relief activities.
- Most students list at least one stress relief activity, the median number of stress relief activities was 3-4.

# EDA & Data Understanding



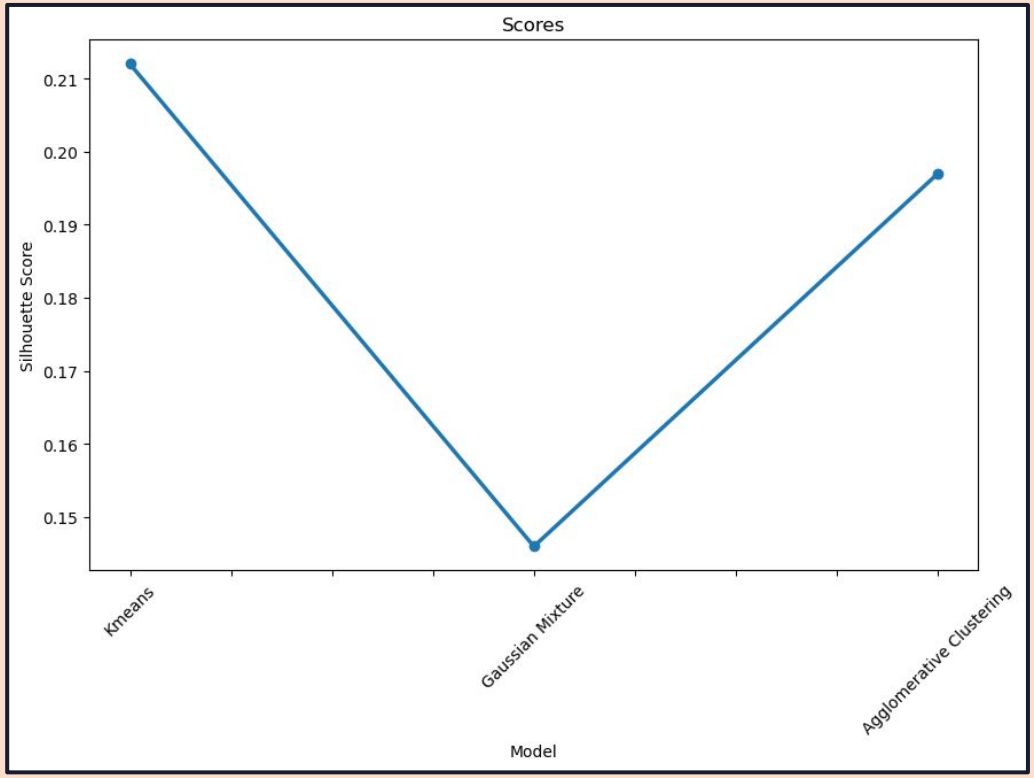



# Initial / Minimum Preprocessing

- Removed student aged 26, which was an outlier.
- Removed the age column, because the range was small and variable ranked low on k-means feature importance chart.
- Kept only Undergraduate students, therefore deleted ‘Post graduates’ then degree_major column.
- Removed 1 student who had no stress relief activity.



# Baseline model



# Modeling Selection

Gaussian Mixture

K-Means

Agglomerative Clustering



K-Means

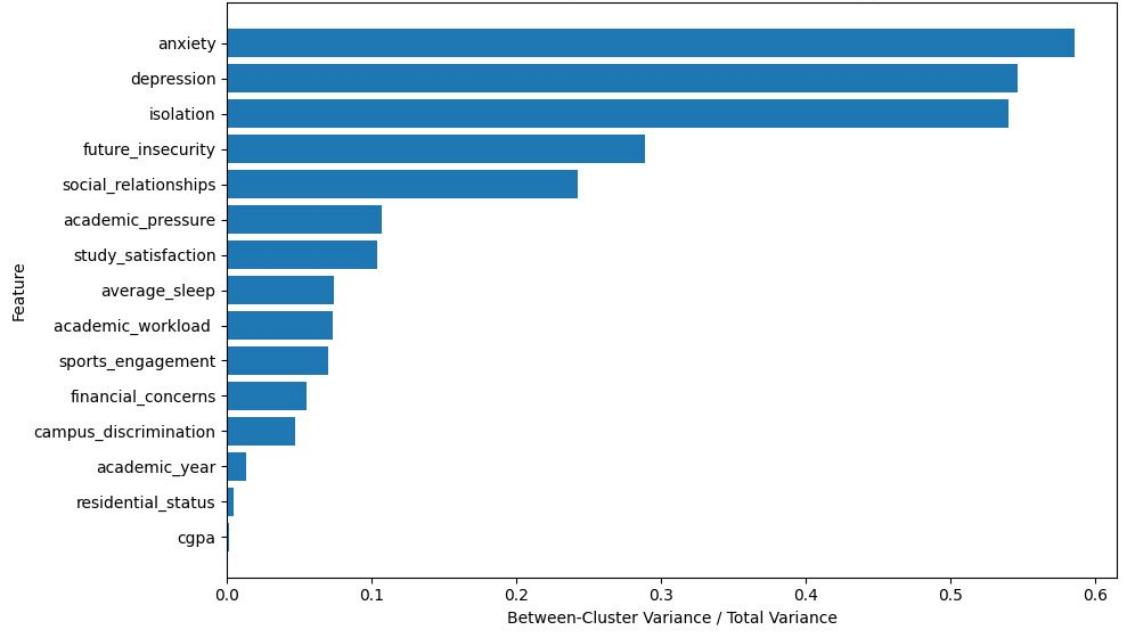


K-Prototype

# Additional Data Preprocessing

| Encoding Technique | Variables |
|---|---|
| Label/Nominal | ‘cgpa’, ‘sports_engagement’, ‘average_sleep’, ‘academic_year’ |
| Binary | ‘gender’, ‘campus_discrimination’, ‘residential_status’ |
| One-hot | ‘degree_major’, ‘university’ |
| Multi-hot | ‘stress_relief_activities’ |
| Min Max Scaling (K-means) | ‘study_satisfaction’, ‘academic_workload’, ‘academic_pressure’, ‘financial_concerns’, ‘social_relationships’, ‘depresion’, ‘anxiety’, ‘isolation’, ‘future_insecurity’ |

# Feature Selection




- **Lowest performing:** *residential_status, cgpa, academic_year.*

- From Cramer's V tests, we know *campus_discrimination* is significantly correlated to *residential_status* therefore we will drop both. As well as *age* & *cgpa.*

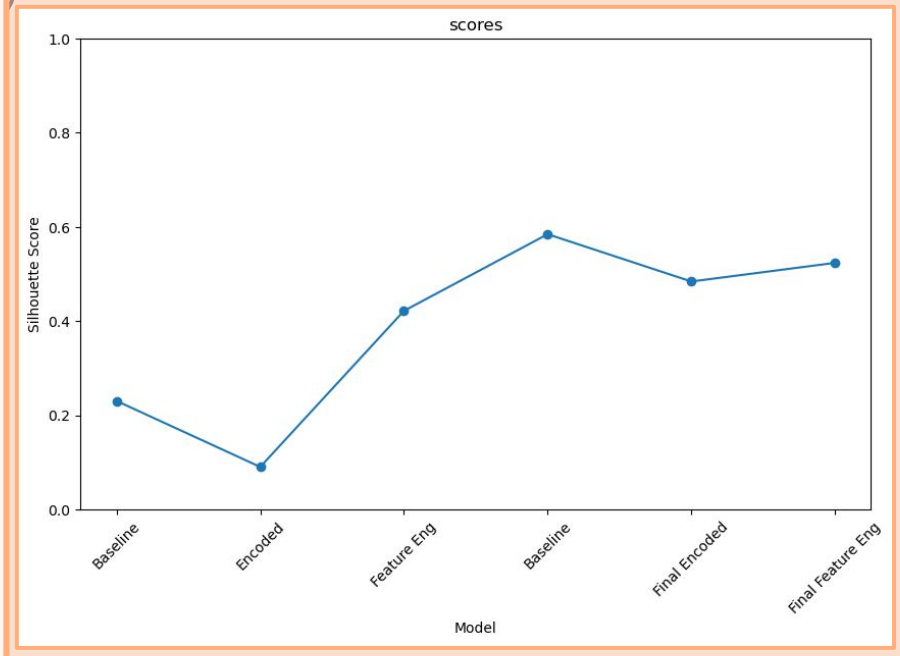
# Feature Engineering

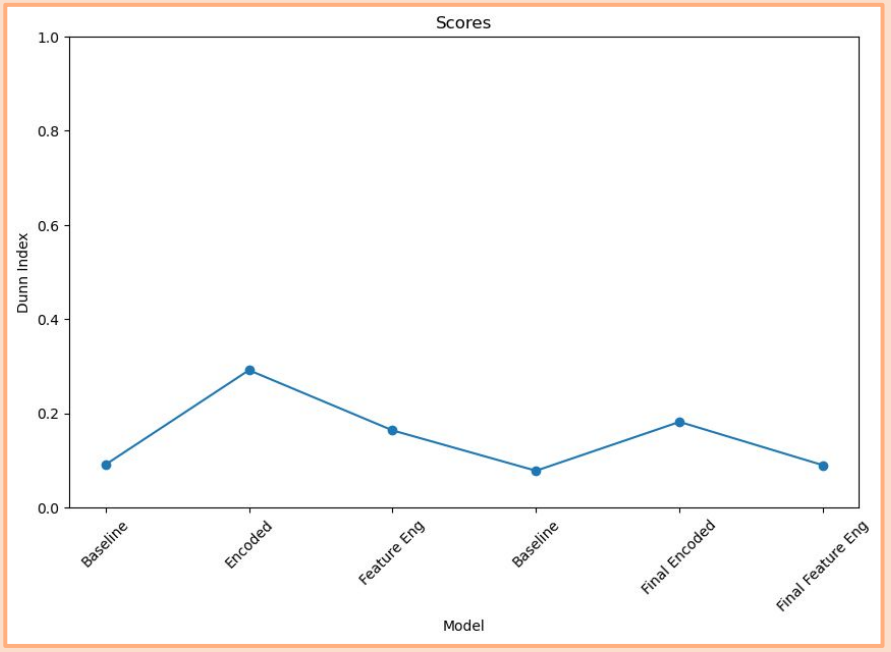
| Original Variable(s) | New Variable(s) | Method |
|---|---|---|
| campus_discrimination, residential_status, age, cgpa | | Removal |
| depression, anxiety | mental_health_score, mental_stressor_score *(includes isolation)* | Score (weighted sum) |
| social_relationships, isolation, sports_engagement | social_iso_ratio, social_sport_ratio | Ratio |
| average_sleep, academic_workload, academic_pressure | sleep_work_ratio, work_pressure_ratio | Ratio |

# Hyperparameter Tuning

| Gamma | Clusters |
|---|---|
| 0.10 | 2 |
| 0.20 | 3 |
| 0.25 | 4 |
| 0.50 | 5 |
| 1.0 | 6 |

# Model Comparison



# Final Model

| K-Prototype |
| --- |
| Encoded |
| Gamma = 0.20 |
| Clusters = 2 |
| Silhouette = 0.48 |
| Dunn = 0.18 |

| | | |
| --- | --- | --- |
| Study satisfaction | 4.18 | 3.44 |
| Academic workload | 3.78 | 4.03 |
| Academic pressure | 3.63 | 4.06 |
| Anxiety | 3.14 | 3.25 |
| Isolation | 3.16 | 3.34 |
| Future insecurity | 2.92 | 3.03 |
| Depression | 3.25 | 3.03 |
| Social relationships | 2.82 | 2.84 |
| Sleeping 7–8 hours | 41% | 28% |
| Sports/fitness as stress relief | 20% | 31% |
| Online entertainment as stress relief | 45% | 34% |

# Limitations

### 1. Limited Dataset

- Relatively small sample size makes generalization difficult
- Limited stress related variables limits reliability of our results

### 2. Sampling & demographics

- Gender imbalance affects our profile representation
- Findings may not generalize to all STEM students in Pakistan

### 3. Cultural Context

- Limited familiarity with local cultural and educational norms
- Stress experiences may vary across regions and universities

### 4. Modeling Challenges

- Unsupervised learning requires careful model selection
- Unfamiliar with correct unsupervised model path

# Conclusion & Future Work

- Used K-prototype clustering to identify two distinct stress profiles among the STEM students
- **Cluster 1:** tends to be less stressed overall, but more depression (less isolation, less academic stress, high satisfaction). More online entertainment and less sports for stress relief.
- **Cluster 2**: tends to be more stressed, less depressed, less sleep. More fitness and less online entertainment.
- Achieved a silhouette score of 0.48.

**Practical Implications**

- Students experience different combinations of academic and personal stressors.
- Universities could use these profiles to better understand student needs and guide support programs.

**1. Expand the dataset**

- Include more students, universities, and regions

**2. Explore demographic differences**

- Investigate gender and other relevant demographic factors

**3. Test the “patient zero” scenario**

- Examine where student with no stress-relief activities would fall within the K-means clusters.



# Thank You!

## Any Questions?

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# Citations

Khan, K. U. D., Gulzar, S., & Yahya, F. (2013). Crucial factors affecting stress: A study among undergraduates in Pakistan. International Journal of Asian Social Science, 3(2), 428–442.

Nasir, P. (2025). Depression, anxiety, and stress among university students in Pakistan. International Journal of Social Sciences, Humanities and Communications, 1(1), 13–26.

Saeed, M., Ullah, Z., & Ahmad, I. (2020). A qualitative exploratory study of the factors causing academic stress in undergraduate students in Pakistan. Liberal Arts and Social Sciences International Journal, 4(1), 203–223.